31/10/26 MLA0205 - Fundamentals of ML

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Assignment - 1

Task 1: Design of Learning Systems

System	Learning Paradigm	Task	E & P
(a) Loan default	Supervised	Classification	E: 10 years of labelled repayment P: Accuracy Precision & recall
(b) Customer segments	Unsupervised	Clustering	E: Customer behaviour data P: Silhouette score
(c) Credit-card spend	Supervised	Regression	E: Historical spending data P: Mean squared Error
(d) ATM cash refill	Reinforcement	Control	E: Repeated trials P: Reward from cash availability
(e) Fraud detection	Semi-supervised	Classification	E: 2% labelled & 98% unlabelled P: Precision & recall

Label noise: System (a) is strongly affected because it depends entirely on labelled repayment records. Incorrect labels can reduce prediction accuracy.

Task 2: Bayes Theorem:

Given

Prevalence = 0.02
Sensitivity = 0.95
Specificity = 0.90
False +ve rate = 0.10

Using Bayes theorem

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|\bar{D})P(\bar{D})}$$

$$= \frac{0.95\ (0.02)}{(0.95)\ (0.02) + (0.10)\ (0.98)}$$

$$= \frac{0.019}{0.117} = 0.1624$$

Answer : 16.24%.

The result is much lower than the sensitivity because the disease is rare.

Task 3: Join Probability Distribution

(a) Margin Probabilities:

$P(X=0) = 0.30$, $P(X=1) = 0.40$

$P(X=2) = 0.30$

$P(Y=0) = 0.45$, $P(Y=1) = 0.55$

(b) Expected values and variance:

$E(X) = 0(0.3) + 1(0.4) + 2(0.3) = 1$

$E(X^2) = 0 + 0.4 + 1.2 = 1.6$

$Var(x) = 1.6 - 1^2 = 0.6$

$E(Y) = 0(0.45) + 1(0.55) = 0.55$

(c) Conditional Probability

$$P(X=2 \mid Y=1) = \frac{0.10}{0.55} = 0.1818$$

(d) Covariance & Correlation

$E(XY) = 0.25 + 0.20 = 0.45$

$Cov(X,Y) = 0.45 - (1)(0.55) = -0.10$

$Var(Y) = 0.55(0.45) = 0.2475$

$$\rho = \frac{-0.10}{\sqrt{0.6 \times 0.2475}} \approx -0.2596$$

(c) Independence

$P(X=0, Y=0) = 0.10$

$P(X=0)\,P(Y=0) = 0.30(0.45) = 0.135$

the values are not equal.

∴ x & y are not independent

Task 4: Normal Distribution & MLE

Given

$X \sim N(50, 16), \sigma = 4$

(a) Probability calculations

For $X > 58$

$$z = \frac{58-50}{4} = 2$$

$P(X > 58) = 1 - \Phi(2) = 1 - 0.9772$

$P(X > 58) = 0.0228$

For $46 < X < 54$

$$z_1 = \frac{46-50}{4} = -1, \quad z_2 = \frac{54-50}{4} = 1$$

$P(-1 < z < 1) = 0.8413 - 0.1587 = 0.6826$

(b) Maximum likelihood Estimates

Mean $\Rightarrow \mu = \frac{12+11+15+14+13}{5} = 13$

Variance $\Rightarrow \sigma^2 = \frac{4+4+4+1+0}{5} = 2$

Unbaised variance $\Rightarrow s^2 = \frac{10}{4} = 2.5$

Task 5: Entropy and Coding

Given Probabilities

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$$

Entropy $H(X) = -\sum P(x) \log_2 P(x)$

$= 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3)$

$= 1.75$ bits

$H(X) \Rightarrow \log_2 4 = 2$ bits

Redundacy: $R = 2 - 1.75 = 0.25$ bits

$\% = \frac{0.25}{2} \times 100 = 12.5\%$

Average code length

code: 111

$L = 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3)$

$L = 1.75$ bits

Since average length equals entropy, the code is optimal.

Task 6 - KL Divergence:

Given $P = (0.5, 0.25, 0.25)$

$Q = (1/3, 1/3, 1/3)$

KL Divergence

$$D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$$

$= 0.5 \log_2(1.5) + 0.25 \log_2(0.75) + 0.25 \log_2(0.75)$

$= 0.08496$ bits

Similarly $D_{KL}(Q||P) = 0.08170$ bits

Conclusion: KL divergence is assymetric. Reversing P and Q gives a different result, so the choice of direction matters when using it as a training loss.

Task 7 - Entropy and Mutual Information

Given $P(0,0) = 0.4$, $P(0,1) = 0.1$

$P(1,0) = 0.1$, $P(1,1) = 0.4$

marginal probabilities of both A & B are (0.5, 0.5)

$H(A) = H(B) = 1$ bit

Joint entropy

$H(A,B) = -2(0.4 \log_2 0.4) - 2(0.1 \log_2 0.1)$

$H(A,B) = 1.7219$ bits

.25 bits

.955 (3)

timal.

onditional entropy:

$$H(A|B) = H(A,B) - H(B)$$
$$= 1.7219 - 1 = 0.7219$$

Mutual information:

$$I(A,B) = H(A) - H(A|B)$$
$$= 0.2781 \text{ bits}$$

Since mutual information is +ve, B provides useful information for predicting A.

Task 8: Bayes Decision Theory:

Given

$$\omega_1 = N(2,4), \quad \omega_2 = N(6,4)$$
$$P(\omega_1) = 0.6, \quad P(\omega_2) = 0.4$$

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{P(\omega_1)}{P(\omega_2)}$$
$$= 4 + \ln(1.5)$$
$$= 4.4055$$

For $x = 4.5$ since $4.5 > 4.4055$ classify it as class ω_2

$$P(\text{error}) = 0.6\,[1 - \Phi(1.2027)] + 0.4\,\Phi(-0.7973)$$

$$P(\text{error}) \approx 0.154$$

∴ Increasing the prior probability of ω_1 moves the threshold to the right, favouring class ω_1

Task 9: Minimum Risk Decision:

Given

$$P(\omega_1|x) = 0.3, \quad P(\omega_2|x) = 0.7$$

loss matrix:

* Declare disease: loss 0 for disease, 8 for healthy
* Declare healthy: loss 50 for disease, 0 for healthy

Risk of declaring disease:

$$R(\alpha_1) = 0(0.3) + 8(0.7) = 5.6$$

Risk of declaring healthy:

$$R(\alpha_2) = 50(0.3) + 0(0.7) = 15$$

Since 5.6 is lesser than 15, the min risk decision declare disease.

Threshold: $8(1-P) < 50P$

$$P > \frac{8}{85} = 0.1379$$

Declare disease when its posterior prob exceeds 13.79

Task 10: Bias - Variance Trade-off

Degree	Training MSE	Validation MSE	Generalisation Gap
1	9	9.5	0.5
2	4.1	4.6	0.5
3	2.2	2.9	0.7
5	0.9	3.6	2.7
9	0.1	11.8	11.7

Analysis:

Degree 1: Underfitting

Degree 3: Best model

Degree 5: Shows overfitting

Degree 9: Overfitting

Task 11 : Bayesian Estimation of coin Bias

Given 7 heads in 10 tosses & a Beta (2,2)

(a) For 10 tosses

ML estimate

$$\hat{\theta}_{ML} = 7/10 = 0.7$$

Beta $(2+7, 2+3)$ = Beta $(9, 5)$

Map estimate

$$\hat{\theta}_{Map} = \frac{9-1}{9+5-2} = \frac{8}{12} = 0.6667$$

$$E(\theta) = \frac{9}{14} = 0.6429$$

(b) For 70 heads in 100 tosses

ML estimate $= \frac{70}{100} = 0.7$

Beta (72, 32)

Map $= \frac{71}{102} = 0.6961$

Posterior mean $= \frac{72}{104} = 0.6923$

As the no. of observations increase, the influence of prior decrease, and the Bayes estimates approach the ML estimate.